

How to read a scientific paper

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Why bother?

- Training of critical faculties
 - You can see whether you agree with conclusions
- Because one day soon you could be writing papers too!



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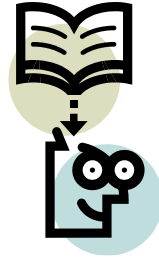
What kind of paper?

- Original research?
- Review, opinion, hypothesis?
- Peer-reviewed?
 - or invitation only
- High-impact journal?
 - author's reputation?

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Why bother?

- Journal papers are current
 - Textbooks are often years out of date
- You can get enough details to replicate what you read about
 - Adapt cutting edge ideas and techniques to your own research

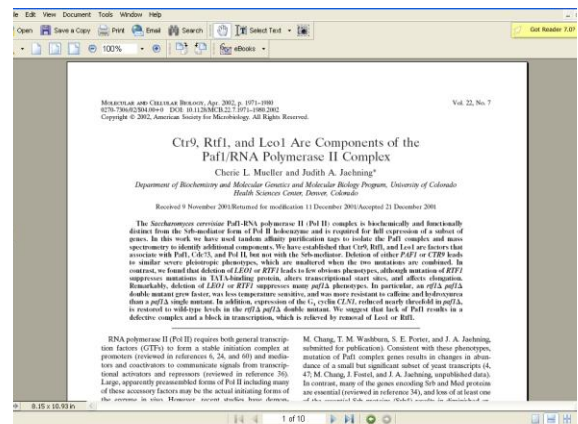


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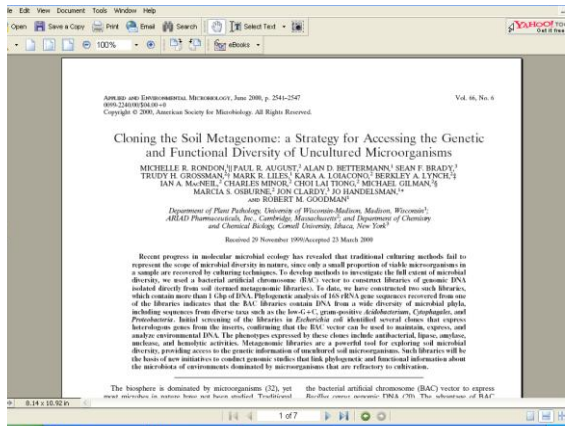
Do I need to read the paper

- For general interest or background information
- To find out exactly what the latest developments are in a field
- To seek evidence to support or refute your ideas
- To broaden your avenues of research
- To find out how a certain piece of research was done

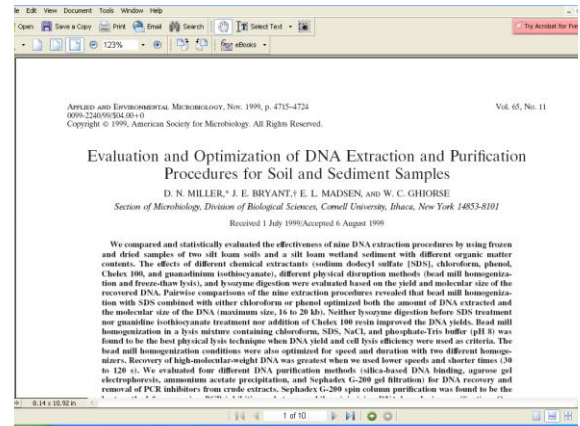
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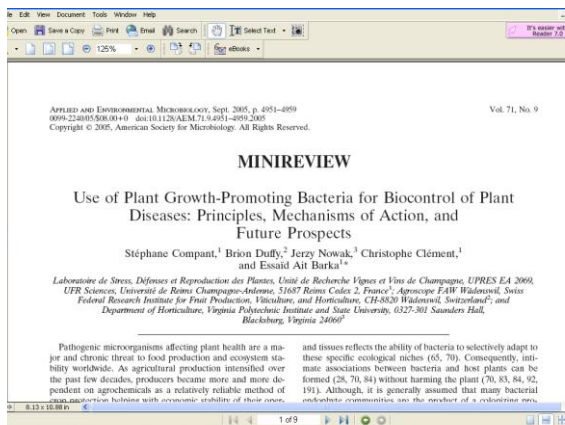
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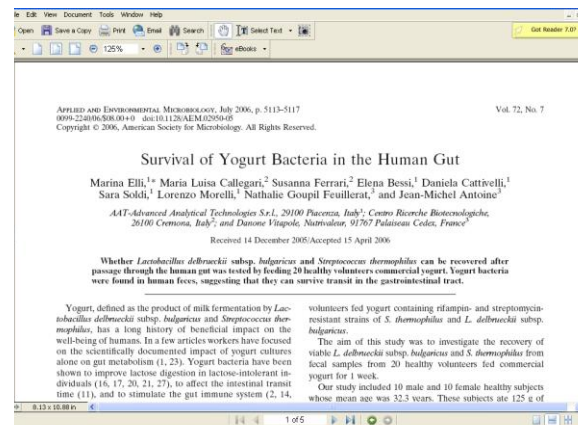
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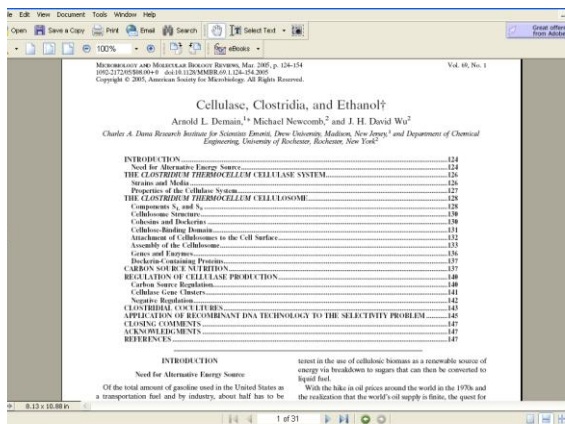
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What kind of paper?

- Papers and journals are judged by their citation rates and impact factors.
- See http://en.wikipedia.org/wiki/Impact_factor
- Also, need to ask is this a specialist journal or general journal?
- Specialist journals in bioinformatics include: Bioinformatics, BMC Bioinformatics, BMC Genomics, Nucleic Acids Research etc
- See <http://www.brc.dcs.gla.ac.uk/~actan/bioinformatics/journals.html>

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Organization of a paper

- IMRAD
 - Introduction, Methods, Results and Discussion
- Plus
 - Title, abstract, authors, acknowledgements, declarations, references
 - Tables and figures; legends

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Organization of a paper

- Variations
 - Pressures on length versus accessibility to non-expert
 - Combined Results and Discussion
 - Methods at end
 - *Science and Nature*
 - On-line supplements

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Reading a scientific paper

- This is not a novel
- No need for a linear approach
- Look at
 - Title
 - Abstract
 - Figures, tables
 - Introduction, results, discussion
 - Then methods

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Reading a scientific paper

- Struggle with the paper
 - active not passive reading
 - use highlighter, underline text, scribble comments or questions on it, make notes
 - if at first you don't understand, read and re-read, spiraling in on central points

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Reading a scientific paper

- Get into question-asking mode
 - doubt everything
 - nit-pick
 - find fault
 - just because it's published, doesn't mean it's right
 - get used to doing peer review



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Reading a scientific paper

- Move beyond the text of the paper
 - talk to other people about it
 - read commentaries
 - consult, dictionaries, textbooks, online links to references, figure legends to clarify things you don't understand



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Blame the authors if...

- Logical connections left out
 - Instead of saying why something was done, the procedure is simply described.
- Cluttered with jargon, acronyms
- Lack of clear road-map through the paper
 - side issues given equal air time with main thread
- Difficulties determining what was done
 - Ambiguous or sketchy description
 - Endless citation trail back to first paper
- Data mixed up with interpretation and speculation

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Why you are reading determines how you should read

- The abstract & introduction should tell you whether it is worth reading in depth or only worth skimming
- The answer will depend on what you are looking for

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Critical assessment of the paper

- Read the experimental results - that is the figures and tables together with their legends - at least as closely as the main text
- Avoid reading the discussion section
- Readers should evaluate results before reading the authors' conclusions
- Use your own judgment

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Evaluating a paper

- What questions does the paper address?
- What are the main conclusions of the paper?
- What evidence supports those conclusions?
- Do the data actually support the conclusions?
- What is the quality of the evidence?
- Why are the conclusions important?

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What questions does the paper address?

- Descriptive research
 - often in early stages of our understanding can't formulate hypotheses until we know what is there.
 - e.g. DNA sequencing and microarray
- Comparative research
 - Ask how general or specific a phenomenon is.
 - e.g. homology searches, comparative genomics

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What questions does the paper address?

- Analytical or hypothesis-driven research
 - test hypotheses
 - e.g. amino-acid composition can be used to predict thermophily
- Methodological research
 - Find out new and better ways of doing things
 - Describe new resources
 - e.g. description of new homology search method, genome database
- Many papers combine all of the above

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What are the main conclusions?

- Do they matter?
 - Of general relevance?
 - Broad in scope?
 - Detailed but with far-reaching conclusions?
 - Accessible to general audience?

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The places to find information about a paper's subject matter

- The title
- The abstract, and
- The introduction

Note

The discussion contains further ideas, but it is not worth reading the discussion in any detail until we have good idea what is being discussed.

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Abstract & Introduction

- Abstract should give you a brief summary of the paper's main finding
- Introduction provide a background to the paper and a rationale for the investigation in more detail than is possible
- The abstract and introduction help you to decide whether, why and how to read

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Readers for their part should approach the abstract with a question in mind : what controversy or orthodoxy does this research take as its starting point ?

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Craig F. Barrett and Matthew A. Parker (2006). *Appl. Environ. Microbiol.* 72(2): 1198-1206.

rRNA gene sequencing and PCR assays indicated that 215 isolates of root nodule bacteria from two *Mimosa* species at three sites in Costa Rica belonged to the genera *Burkholderia*, *Cupriavidus*, and *Rhizobium*. This is the first report of *Cupriavidus* sp. nodule symbionts for *Mimosa* populations within their native geographic range in the neotropics. *Burkholderia* spp. predominated among samples from *Mimosa pigra* (86% of isolates), while there was a more even distribution of *Cupriavidus*, *Burkholderia*, and *Rhizobium* spp. on *Mimosa pudica* (38, 37, and 25% of isolates, respectively). All *Cupriavidus* and *Burkholderia* genotypes tested formed root nodules and fixed nitrogen on both *M. pigra* and *M. pudica*, and sequencing of rRNA genes in strains reisolated from nodules verified identity with inoculant strains. Inoculation tests further indicated that both *Cupriavidus* and *Burkholderia* spp. resulted in significantly higher plant growth and nodule nitrogenase activity (as measured by acetylene reduction assays) relative to plant performance with strains of *Rhizobium*. Given the prevalence of *Burkholderia* and *Cupriavidus* spp. on these *Mimosa* legumes and the widespread distribution of these plants both within and outside the neotropics, it is likely that both β -proteobacterial genera are more ubiquitous as root nodule symbionts than previously believed.

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Why it is good idea to read introductions

- They give you some idea what background information you need before starting
- They give you an insight into the authors' starting point and approach to the subject

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Until 2001, all bacteria known to be involved in root nodulesymbioses with legume plants were restricted to genera within the α -Proteobacteria (*Rhizobium*, *Sinorhizobium*, *Mesorhizobium*, *Bradyrhizobium*, and *Azorhizobium*) (37). This changed when Moulin et al. (14) discovered two nodule-forming isolates of the β -proteobacterial genus *Burkholderia* on legumes in Africa and South America. They suggested the terms α - and β -rhizobia to distinguish these two phylogenetic lineages of nodule-symbiotic *Proteobacteria*. Members of two other genera within the β -Proteobacteria are now known to be legume nodule symbionts. Chen et al. (3) described the novel species *Ralstonia taiwanensis* as a symbiont of *Mimosa pudica* in Taiwan. This species was subsequently transferred to the genus *Cupriavidus* (31). In a study across 14 sites in Taiwan, *Cupriavidus taiwanensis* was found to be the dominant symbiont associated with the legumes *Mimosa pudica* and *Mimosa diplotricha*, and isolates of *Burkholderia caribensis* also occurred as nodule symbionts in this region (4). Both *M. pudica* and *M. diplotricha* are plants endemic to the neotropics that have been naturalized in Taiwan (1, 4, 11, 36). Another recently described β -proteobacterium (*Herbaspirillum lusitanum*) was found in Portugal to nodulate *Phaseolus vulgaris* (28).

Data are still limited regarding the symbiotic relationships of rhizobia and mimosoid legumes in their native geographical range.

The title of the paper

Coexistence of *Burkholderia*, *Cupriavidus*, and *Rhizobium* sp. Nodule Bacteria on two *Mimosa* spp. in Costa Rica

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Summary

- The abstract and introduction should explain why the paper was written
- They do not give detailed information, but should help you decide how much time to spend on the paper
- Introductory sections are an entry into a paper - never substitute for reading it properly

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What evidence supports them?

- Look at Results section and relevant tables and figures.
 - May be one primary experiment to support a point.
 - More often several different experiments or approaches combine to support a particular conclusion.
 - First experiment might have several possible interpretations, and the later ones are designed to distinguish among these.
- In the ideal case, the Discussion begins with a section of the form "Three lines of evidence provide support for the conclusion that...."

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Judging the quality of the evidence

- You need to understand the methods thoroughly
 - may need to consult textbooks
- You need to know the limits of the methods
 - e.g. an assignment of distant homology has to be treated as working hypothesis rather than fact
- Separate fact from interpretation
- Are the results expected?
 - Extraordinary claims require extraordinary evidence

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Judging methods

- There has to be a logical reason why the method can or may answer the question
- Defined and reproducible protocols must be followed
- Controls must be in place in order to rule out extraneous influences on the results

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Judging the quality of the evidence

- Look at details, assess them for plausibility
 - The veracity of whole depends on the veracity of its parts!
 - e.g. look at gene lists, what is missing but expected, what is present, but unexpected?
- Where are the controls?
- What is the gold standard?
 - e.g. when predicting protein-coding genes, when evaluating annotation, how can you assess accuracy?

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Why it is good idea to read materials and methods

- To know how it was done in order to understand what it means
- If you want to replicate an experiment, the methods section is indispensable
- To find stimulating ideas and make connections between different areas
- To adapt methodological approaches to our own experiments

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Do the data support the conclusions?

- Data may be believable but not support the conclusion the authors wish to reach
 - logical connection between the data and the interpretation is not sound (often hidden by bad writing)
 - might be other interpretations that are consistent with the data

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Do the data support the conclusions?

- Rule of thumb
 - If multiple approaches, multiple lines of evidence, from different directions, supporting the conclusions, then more credible.
- Question assumptions!
 - Identify any implicit or hidden assumptions used by the authors in interpreting their data?

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Conclusion

Peer review: you are the judge!

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